

FACTSHEET FS05 · Monkey portfolios

Can a blindfolded monkey beat the index?

Henrik Gjerner · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · data to 31 August 2026

STRATEGY	ORIGIN	FAMILY
Random stock picks, equal weight, rebalanced yearly	Malkiel (1973), <i>A Random Walk Down Wall Street</i>	Portfolio folklore
UNIVERSES	BACKTEST	BENCHMARK
US · EU · UK · DK · SCANDI · World (point-in-time)	Feb 2013 – Aug 2026, 10,000 monkeys per market	Size-weighted universe ("the index") and equal-weight universe
REBALANCE	STATUS	
Every January	Descriptive factsheet, not a registered trial	

The claim. Burton Malkiel (1973): a blindfolded monkey throwing darts at the stock pages could pick a portfolio that does as well as the experts. Arnott, Hsu, Kalesnik & Tindall (2013) found that random, and even 'upside-down', portfolios beat the cap-weighted US index over 1964–2012. **What we find:** the share of 10,000 monkeys that beat the index ranged from 16% to 86% across the six markets: US 16%, EU 71%, UK 85%, DK 85%, SCANDI 86%, World 72%. Measured by Sharpe ratio, the same or fewer monkeys win (US 3%, EU 64%, UK 64%, DK 85%, SCANDI 77%, World 57% beat the index's Sharpe). **What decides it (§10):** whether small beats big that year. The yearly share of winning monkeys moves with the gap between the equal-weight universe and the index (correlation 0.56 to 0.94). The average monkey is the equal-weight universe; the typical (median) monkey lags it by the cost of holding only 30 names. **Classification (§11):** a benchmark lesson, not a strategy: a random portfolio is an equal-weight portfolio, and equal weight is a size tilt.

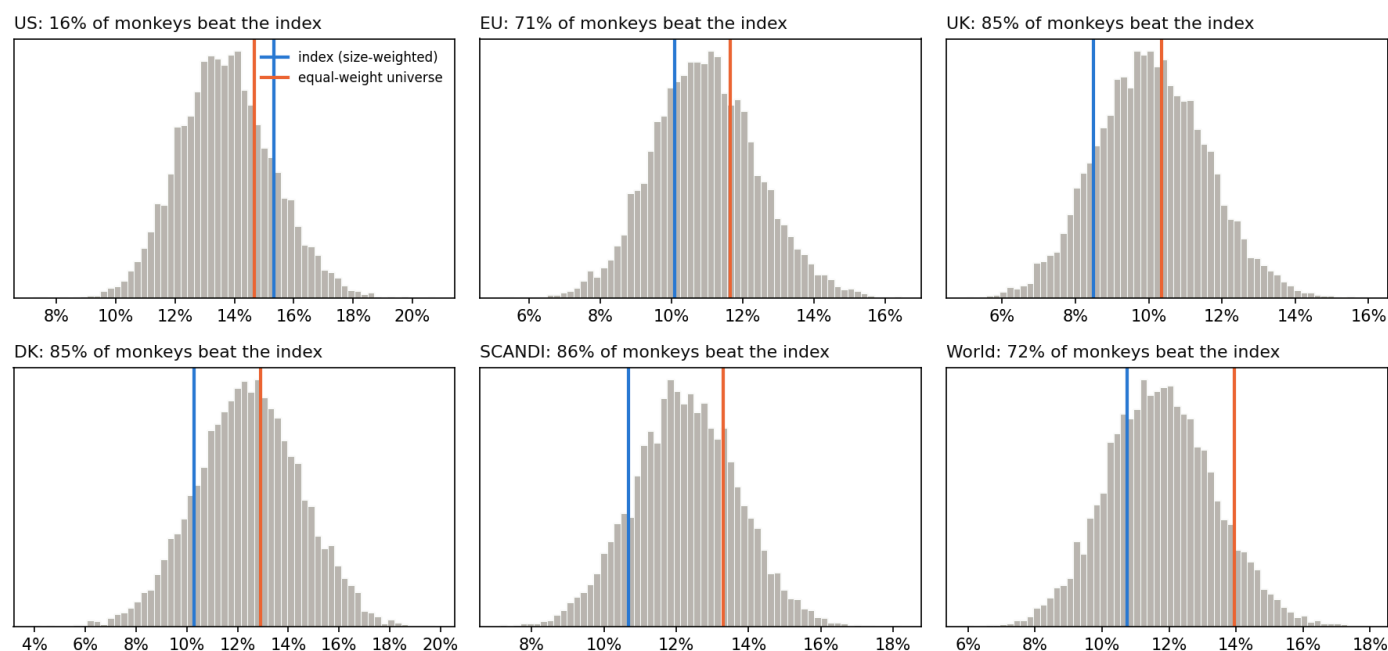
1. Headline performance

Universe	Stocks per monkey	Median monkey CAGR	5th–95th percentile	Average monkey CAGR	Index CAGR	EW universe CAGR	Monkeys beating the index (return)	Beating on Sharpe	Median Sharpe monkey / index	Max DD median monkey / index
US	30	+13.7%	+11.2% to +16.5%	+13.8%	+15.3%	+14.7%	16%	3%	0.90 / 1.07	-25% / -24%
EU	30	+10.9%	+8.6% to +13.5%	+11.0%	+10.1%	+11.6%	71%	64%	0.78 / 0.75	-27% / -26%
UK	30	+10.1%	+7.7% to +12.6%	+10.2%	+8.5%	+10.4%	85%	64%	0.74 / 0.71	-30% / -27%
DK	10	+12.4%	+9.0% to +15.8%	+12.6%	+10.3%	+12.9%	85%	85%	0.79 / 0.68	-33% / -40%
SCANDI	20	+12.3%	+9.9% to +14.6%	+12.4%	+10.7%	+13.3%	86%	77%	0.87 / 0.80	-29% / -27%

Universe	Stocks per monkey	Median monkey CAGR	5th–95th percentile	Average monkey CAGR	Index CAGR	EW universe CAGR	Monkeys beating the index (return)	Beating on Sharpe	Median Sharpe monkey / index	Max DD median monkey / index
World	30	+11.7%	+9.2% to +14.5%	+11.9%	+10.8%	+13.9%	72%	57%	0.78 / 0.76	-30% / -28%

Index = size-weighted universe: market-cap weights in the US (Sharadar), traded-value weights elsewhere (a proxy; no shares outstanding outside the US). 10 bp in and out each January.

10,000 dartboard portfolios per market: annual return 2013–2026



2. Strategy description

Malkiel's line in *A Random Walk Down Wall Street* (1973) became a test. The Wall Street Journal ran a Dartboard contest from 1988 to 2002 in which professionals' picks were set against darts; the pros won more often, but part of their edge was the publicity of being named in the paper (Barber & Loeffler 1993; Metcalf & Malkiel 1994). Arnott, Hsu, Kalesnik & Tindall (2013) showed why random portfolios tend to beat cap-weighted indices: any weighting that ignores market cap tilts towards smaller and cheaper stocks.

Rule	Original	This factsheet
Selection	Darts at the stock pages	Uniform random draw from the point-in-time universe
Size of portfolio	30 stocks in Arnott et al.	30 (US, EU, UK, World), 20 (SCANDI), 10 (DK)
Weighting and holding	Equal weight, rebalanced yearly	Same; buy and hold within the year
Benchmark	Cap-weighted index	Size-weighted universe (see note) and equal-weight universe

3. Data load

Universe	Prices	Membership	Names eligible (median)	Currency
US	Sharadar SEP <code>closeadj</code> (total return), 805 tickers incl. delisted	Point-in-time: top 500 by market cap each month-end (Sharadar filings × price, A06 method)	~499	USD
EU	Project1 yfinance cache, Adj Close (TR)	Point-in-time: union of 11 national blue-chip indices (CAC 40, DAX, AEX, IBEX 35, FTSE MIB, OMXS30, OMXC25, OMXH25, BEL 20, PSI-20, WIG20), quarterly Wikipedia snapshots 2012–2026; 83% of member-quarters priced	~238	local (EUR, SEK, DKK, PLN)
UK	Project1 cache, Close + pence dividends (Yahoo under-adjusts LSE ~100×)	Point-in-time: FTSE 100 + FTSE 250, quarterly snapshots; 75% priced	~242	GBP
DK	Project1 cache, Adj Close	Point-in-time: OMX Copenhagen 25, quarterly snapshots; 91% priced	~19	DKK
SCANDI	Project1 cache, Adj Close	Point-in-time: OMXC25 + OMXS30 + OMXH25 (Oslo has no snapshot history); 88% priced	~62	local (DKK, SEK, EUR)
World	US Sharadar + UK + EU above	Point-in-time union of the three; local-currency returns summed, not converted	~963	mixed

Membership comes from Project2's cache of dated Wikipedia index pages (the same parsing as `run_wiki_pit.py` / `run_eu_pit.py`), with a short list of verified ticker renames (e.g. `NZYM-B` → `NSIS-B`, `WDH` → `DEMANT`, `DAI` → `MBG`). A name is eligible from the day after the snapshot that lists it, once it has 60 days of prices. Member-quarters without a price are mostly delisted names that Yahoo no longer serves, so some survivorship remains; see §13.

- **Data gap:** no market caps outside the US; the index is proxied by traded-value weights, which over-weight the most traded (not necessarily largest) stocks.

4. "Signal" creation

None: each January 10,000 monkeys draw their stocks uniformly at random from the names eligible at the end of December (seeded, reproducible).

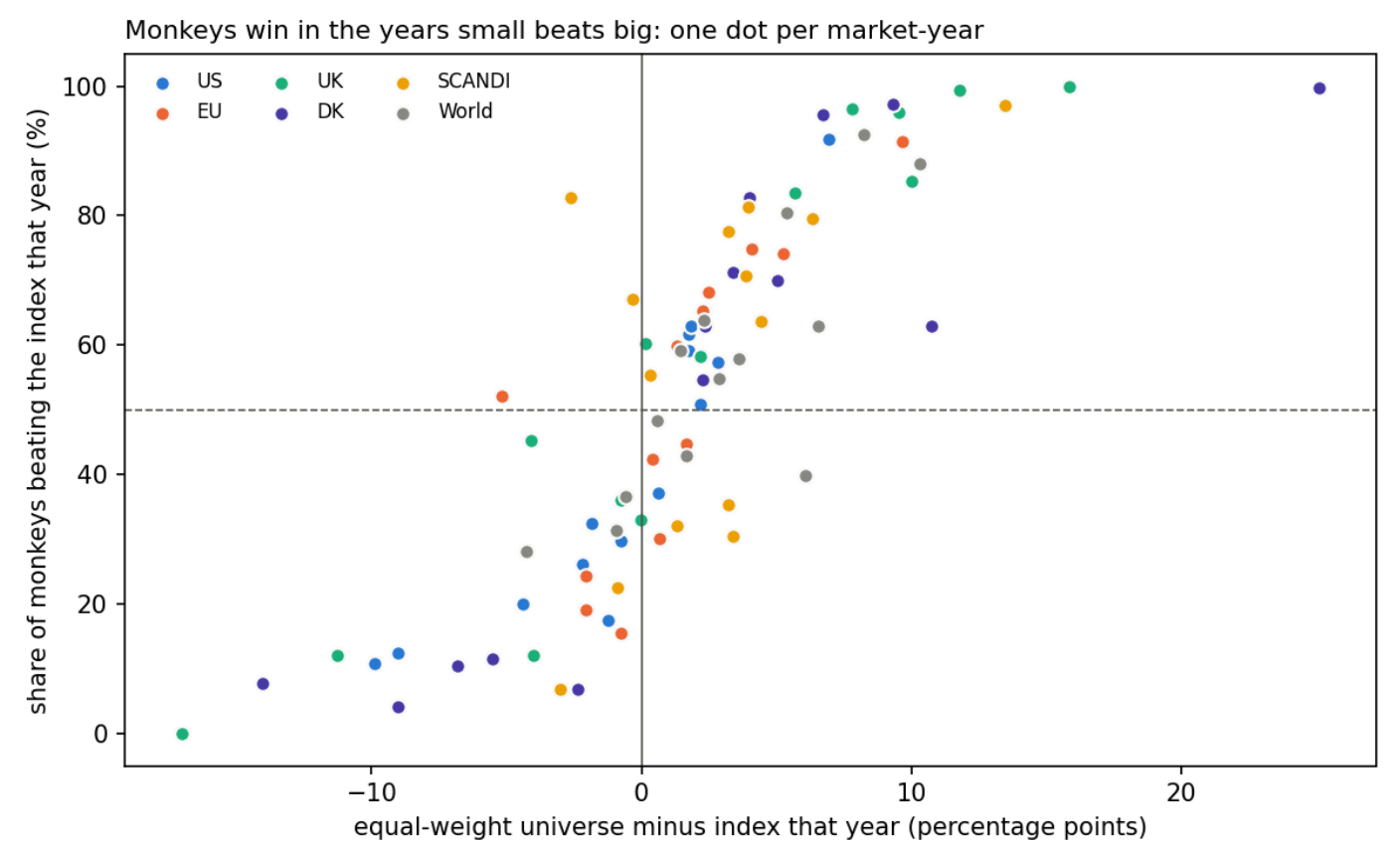
5. Model build

Equal weight at formation, buy and hold for the calendar year; a name that stops trading earns zero after its last price. 20 bp round trip charged each January. Code: `code/monkey.py`, `code/build_monkey.py`.

6. Performance in detail

Share of monkeys beating the index each year (equal-weight universe minus index in brackets)

Year	US	EU	UK	DK	SCANDI	World
2013	59% (+2%)	75% (+4%)	99% (+12%)	63% (+11%)	79% (+6%)	88% (+10%)
2014	62% (+2%)	19% (-2%)	96% (+8%)	7% (-2%)	35% (+3%)	92% (+8%)
2015	32% (-2%)	65% (+2%)	100% (+16%)	8% (-14%)	55% (+0%)	80% (+5%)
2016	63% (+2%)	52% (-5%)	45% (-4%)	97% (+9%)	83% (-3%)	59% (+1%)
2017	30% (-1%)	60% (+1%)	96% (+10%)	71% (+3%)	77% (+3%)	55% (+3%)
2018	26% (-2%)	65% (+2%)	58% (+2%)	83% (+4%)	81% (+4%)	64% (+2%)
2019	37% (+1%)	24% (-2%)	83% (+6%)	12% (-6%)	71% (+4%)	58% (+4%)
2020	17% (-1%)	15% (-1%)	85% (+10%)	55% (+2%)	64% (+4%)	40% (+6%)
2021	51% (+2%)	45% (+2%)	36% (-1%)	63% (+2%)	32% (+1%)	63% (+7%)
2022	92% (+7%)	74% (+5%)	0% (-17%)	4% (-9%)	22% (-1%)	48% (+1%)
2023	11% (-10%)	42% (+0%)	60% (+0%)	11% (-7%)	7% (-3%)	31% (-1%)
2024	12% (-9%)	68% (+2%)	12% (-4%)	96% (+7%)	67% (-0%)	37% (-1%)
2025	20% (-4%)	91% (+10%)	12% (-11%)	100% (+25%)	97% (+13%)	28% (-4%)
2026	57% (+3%)	30% (+1%)	33% (-0%)	70% (+5%)	31% (+3%)	43% (+2%)



7. Trading record

Each monkey trades once a year. The average of all monkeys equals the equal-weight universe held from January (differences come from mid-year index changes and costs). The median monkey lags the average (US -0.2%, EU -0.1%, UK -0.1%, DK -0.2%, SCANDI -0.1%, World -0.2% a year) because a concentrated portfolio compounds below its average return: the more volatile the stocks, the larger the gap.

8. Risk and attribution

Median monkey minus index, regressed on the equal-weight universe minus index (monthly, Newey-West t):

Universe	Slope on EW – index	Alpha / yr	t	Correlation of EW – index with the small-minus-big size sort
US	0.94	-1.2%	-3.49	+0.69
EU	0.80	-0.7%	-1.24	+0.71
UK	0.92	-0.4%	-0.96	+0.80
DK	1.00	+0.1%	0.19	+0.61
SCANDI	0.70	-0.4%	-0.43	+0.57
World	0.74	-1.5%	-3.56	+0.52

The slope near 1 says the monkey is the equal-weight universe; what is left (the alpha) is mostly the concentration drag from holding few names.

9. Statistical verdict

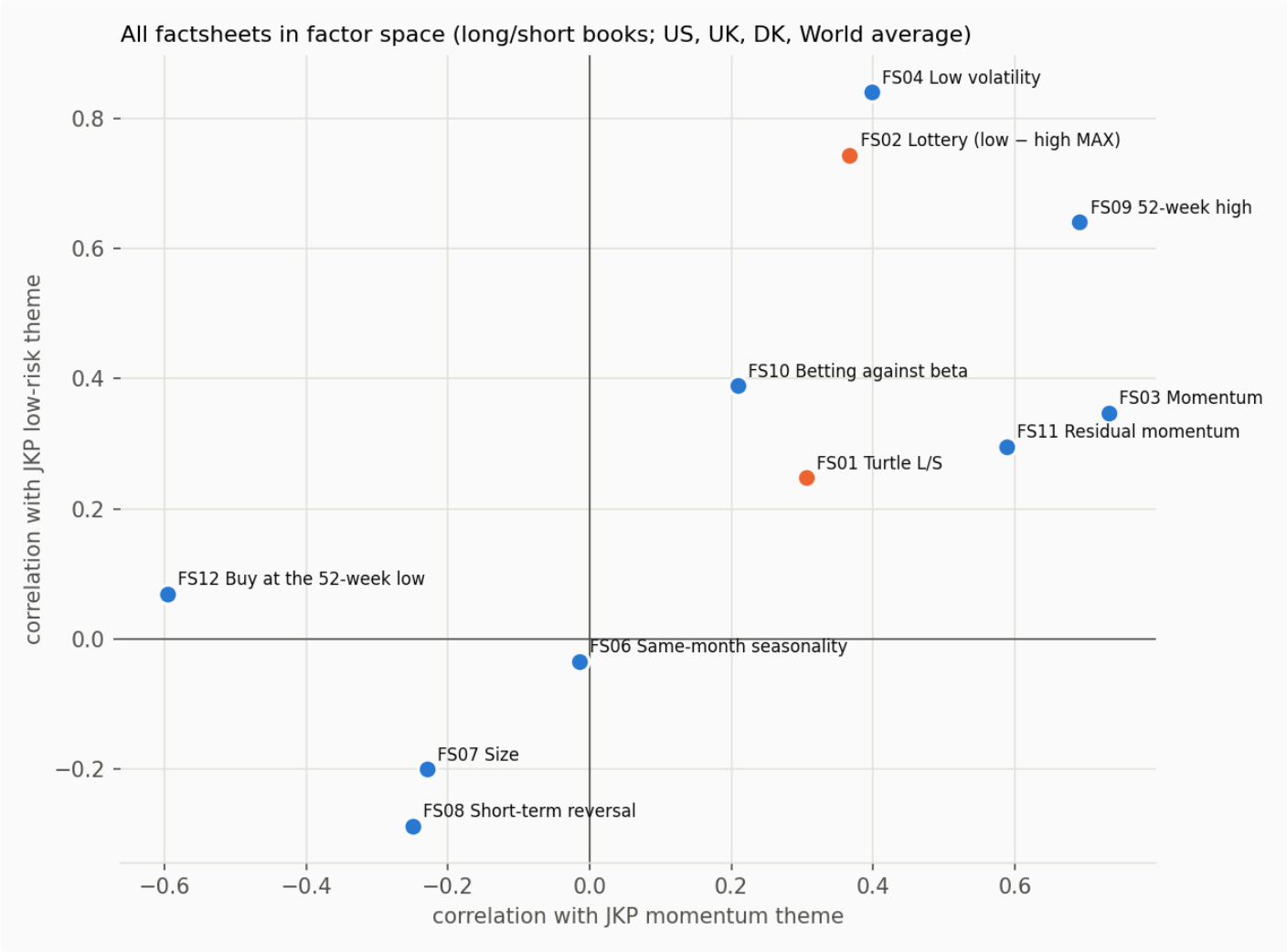
- No hypothesis test is needed for the headline: the monkeys' expected return equals the equal-weight universe by construction. The question is only how equal weight compares with size weight in each market, and that is the size effect (FS07).
- The share beating the index is above 50% in 5 of 6 markets, which are the markets where the equal-weight universe beat the index over the period.

10. What goes wrong, and how to fix it

Nothing goes wrong: the monkey does exactly what equal weighting does. It wins when small stocks beat big ones (UK, DK, EU over this period) and loses when mega-caps lead (the US). Two lessons for any stock-picking strategy: (1) **always compare with the equal-weight universe as well as the index**, otherwise a size tilt looks like skill; (2) **concentration costs**: the typical 30-stock portfolio compounds below the average one. The "fix" is simply to hold more names, which moves the monkey towards the equal-weight universe.

11. Classification: where does it fit?

Dimension	Monkey portfolios
Series family	Portfolio folklore
Academic style	Benchmark test; non-cap weighting (Arnott et al. 2013)
Signal	None (random)
Exposure	Equal-weight minus cap-weight: small-cap tilt (and value, per Arnott et al.)
Where it fits	A benchmark , not a factor. It belongs in the factor battery only as the equal-weight reference against which every other strategy is judged



12. 360° view

The monkey's neighbourhood is the equal-weight universe (identical in expectation) and the size sort: the equal-weight-minus-index spread correlates US +0.69, EU +0.71, UK +0.80, DK +0.61, SCANDI +0.57, World +0.52 with the small-minus-big size sort of the signal library. Everything the monkey "knows" is size.

13. Caveats

1. **Index proxy.** Outside the US the index is traded-value weighted, which differs from free-float cap weights.
2. **Residual survivorship.** 9–25% of member-quarters outside the US lack a price; delisted names are the most likely missing.
3. **Currency.** Local currency; World mixes currencies.
4. **Not a registered trial.**

14. Academic references

- Malkiel, B. (1973). *A Random Walk Down Wall Street*. W. W. Norton.
- Arnott, R., Hsu, J., Kalesnik, V. & Tindall, P. (2013). The surprising alpha from Malkiel's monkey and upside-down strategies. *Journal of Portfolio Management*, 39(4), 91–105.
- Barber, B. & Loeffler, D. (1993). The "Dartboard" column: Second-hand information and price pressure. *Journal of Financial and Quantitative Analysis*, 28(2), 273–284.
- Metcalf, G. & Malkiel, B. (1994). The Wall Street Journal contests: The experts, the darts, and the efficient market hypothesis. *Applied Financial Economics*, 4(5), 371–374.
- Banz, R. (1981). The relationship between return and market value of common stocks. *Journal of Financial Economics*, 9(1), 3–18.

15. Reproduce

`code/xs.py` (inputs) → `monkey.py` → `build_monkey.py`.